

MAE 2250 - ODP 6: Client Report

Neil Morrison, Susanna Aufrechtig, Jamie Dalvito, Flavia Capet, and Luca Welle

Context and Problem Statement

Spotted Lanternflies (SLFs), an invasive species rapidly spreading across the eastern United States, pose a growing threat to agricultural systems.² SLFs land on grape vines and feed on plant sugars, contaminating harvests, reducing yields, and worsening grapes.² SLFs also excrete honeydew, promoting growth of sooty mold that inhibits photosynthesis.⁵ These effects can cost the wine industry billions of dollars in lost yields.³ On-vine SLF removal was shown to be impractical because the insects are numerous, and aggressive removal methods could damage the vines.

Post harvest removal proved to be too risky considering the high rate of contamination. So instead, we wondered if there was a way to attract SLFs away from the vines completely.



Final Prototype and Application

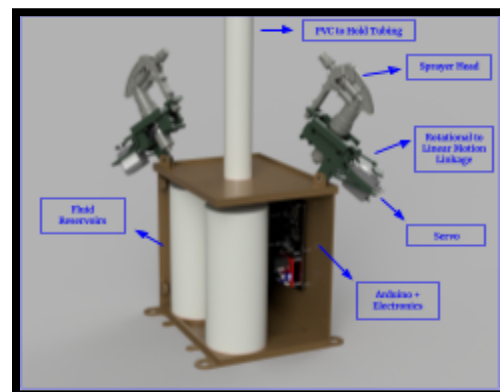
To lure SLFs away from grapevines, our system utilizes a sugary attractant designed to mimic the sap of the Tree of Heaven, a preferred host plant⁵. Once lured to the device, SLFs are then exterminated using targeted vinegar¹ spraying through side-mounted sprayers.

1. Sap Attraction: We pump a sugar solution that mimics the viscosity of natural sap through a central pole, creating an attractive feeding source.

2. Targeted Elimination: Once the insects gather, we spray them with vinegar using servo-actuated sprayers.

Key features:

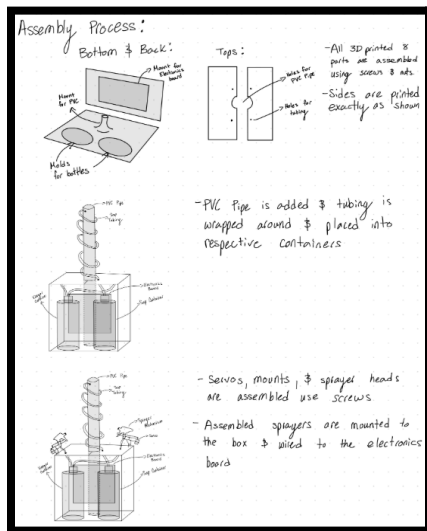
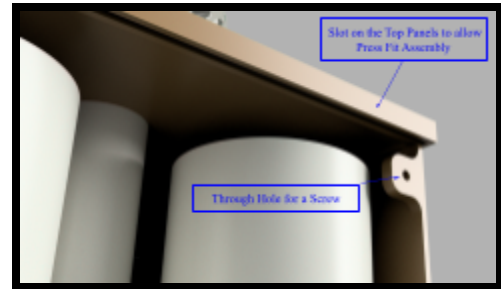
- (1) Adjustable spray angles for coverage optimization
- (2) Compact footprint for vineyard integration
- (3) Arduino-controlled automation
- (4) Low-toxicity vinegar-based treatment



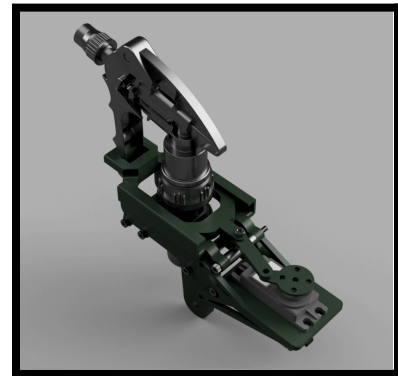
A vineyard simply places the device, turns it on, and it runs autonomously with minimal maintenance.

Assembly

The system is designed so the box exterior can be rapidly assembled or disassembled if need be. Each panel contains a lip to maintain box geometry. Screws are then attached throughout each panel for rigidity. The lid of the box consists of two pieces which are loosely pressed to fit to the top of the box, giving quick access to the interior electronics and fluid reservoirs for maintenance. One of the panels contains a mounting profile for the electrical board which is held on with four screws on



the interior of the box. Electrical components are wired together using jumper wires and Wago



connectors which can be plugged directly into the arduino and allow for easily connecting components in parallel. A 3D printed linkage allows servo motors to actuate the sprayer heads. A 3D printed bar is screwed into the servo horn which pulls the actuator forwards and backwards. Everything is then rigidly attached together using screws and mounted on the top box panel.

Testing Details and Results

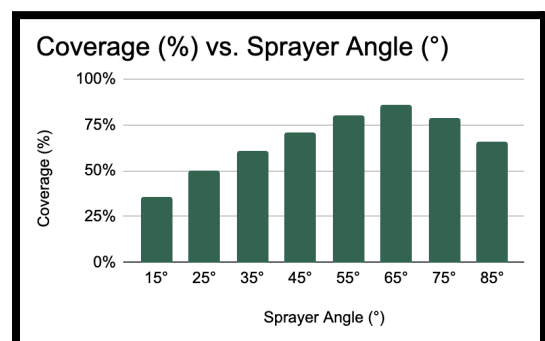
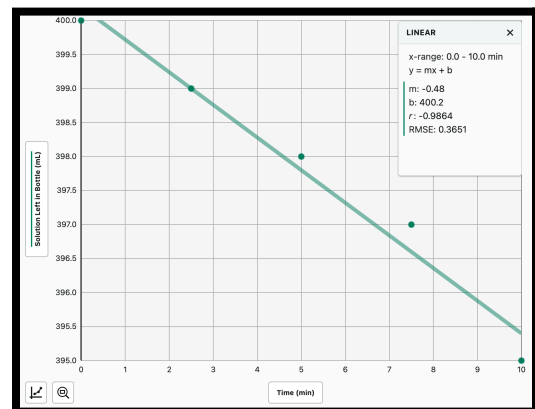
- Sap Drainage Test:** To test to longevity of our sap reservoir, we conducted a drainage test to determine how long the system could realistically last in the field without being refilled

Procedure:

- Mix 200 g sucrose + 300 g water to mimic Tree of Heaven sap viscosity ($\sim 5 \text{ mPa}\cdot\text{s}$)⁴
- Pump solution for 10 min, recording water level every 2.5 min

Results:

- Drainage rate: 0.5 mL/min (6mL/day) $\rightarrow$ 1000mL reservoirs must be refilled every $\sim 167 \text{ da}^2$. **Spray**



Angle & Coverage: To confirm that our sprayers could cover the majority of the central pole in vinegar, we tested sprayer angles to find max coverage.

Procedure:

- (1) Cover central pole in paper, set sprayers to a specific angle, perform a spraying cycle, measure area of paper that is damp

Results:

- The optimal working angle of the sprayers was found to be 65 degrees, achieving 86% spray coverage. This allows for most coverage of the landing area of the SLF on our system

3. Battery Life

To evaluate VineGuard's power constraints, we calculated energy consumption.

Results:

- VineGuard can operate for approximately 8.2 days on a 10,000mAh battery charge. The system's modular battery design enables users to balance cost against serviceability depending on deployment needs.

Power Consumption & Battery Life					
Component	Current Draw (Active) (mA)	Active Time (hr/day)	Current Draw (Idle) (mA)	Idle Time (hr/day)	Daily Consumption (mAh/day)
Pump	300	0.2	0	0	60
Servo	800	0.006	8	23.994	200.9
Arduino	40	24	—	—	960
Total	—	—	—	—	1220.9

Battery Performance	
Parameter	Value
Battery Capacity	10,000 mAh
Daily Usage	1220.9 mAh
Battery Life	~8.2 days

Conclusion and Recommendation

Based on our test results, Vineguard demonstrates a strong potential as a feasible and effective solution for protection against SFLs. Our testing showed that the system can deliver the attractant at controlled flow rates, achieve ideal spray coverage under specific angles, and operate for around 167 days completely autonomously based on power supply. Additionally, the manufacturing and assembly process required minimal setup, and the system showed low maintenance demands once built. Given this performance we would recommend development of our prototype to move forward with field testing as the next step. It will be important to test the attractant that will be used, along with ideal placement that will bring in the most insects at a time. Also, taking tourism into account is important so the aesthetics of the design would also need to be modified to camouflage it into the vineyard as much as possible. Overall, Vineguard offers a feasible low-toxicity, low-intervention solution for mitigating SLF damage in vineyards through a bio-inspired decoy system. This prototype has the potential to not only save vineyards millions in yields but also to provide an eco-friendly, passive, low maintenance solution compared to past pest control methods.

BOM for Final Prototype:

Description	Vendor	McMaster Code	Quantity	Unit of measurement	Total Cost
Soft Masterkleer PVC Tubing for Air and Water	McMaster Carr	5233K113	1	25 feet	\$11.50
Push to Connect Fitting	McMaster Carr	3619N12	6	Pack of 1 Each	\$15.66
Kamoer NKP low flow peristaltic pump	Amazon		1	Pack of 1	\$9.98
M3 x 0.5 Black-Oxide Alloy Steel Socket Head Screw 14mm Long	McMaster Carr	91290A119	1	Pack of 1	\$15.30
12V Power Supply	Taylor Design Studio		1		~
Metal Servo Arms Horn Aluminum	Amazon		1	Pack of 6	\$6.99
Arduino Uno REV3	Amazon		1	Pack of 1	\$27.60
L298N Motor Driver	Amazon		1	Pack of 2	\$6.98
LM2596 DC to DC Buck Converter	Amazon		1	Pack of 5	\$7.99
Wago 221-415 Lever-Nuts	Amazon		1	Pack of 10	\$9.85
Round Rocker Switches	Amazon		1	Pack of 5	\$6.39
Spray Bottle Long-Reach, 1 Gallon Capacity	McMaster Carr	9864T16	2	Pack of 1	\$8.80
Routing Clamp 304 Stainless Steel, 2 Mounting Points, 15/16" ID	McMaster Carr	8874T43	2	Pack of 1	\$7.42
Semi-Clear HDPE Plastic Bottle 32 FL. oz./1000 ml Capacity, 1-1/2" Mouth OD	McMaster Carr	3681T77	2	Pack of 1	\$7.46
2000 Series 5-Turn, Dual Mode Servo	Taylor Design Studio		1		~
Black-Oxide Alloy Steel Socket Head Screw M3 x 0.5 mm Thread Size, 40 mm Long	McMaster Carr		1	Pack of 25	\$4.86
Zinc-Plated Steel Hex Nut Medium-Strength, ISO Class 8, M3 x 0.5mm Thread Size	McMaster Carr	90591A250	1	Pack of 100	\$3.06
Total Cost					\$149.84

Works Cited

- [1] Amdro, “How to control and kill spotted lanternflies.” Available: Amdro website.
- [2] CNBC, “Spotted lanternflies are feasting on U.S. grapevines and putting vineyards at risk,” Oct. 13, 2022.
- [3] Cornell Chronicle, “Spotted lanternflies could cost NYS grape industry millions,” Jan. 27, 2025.
- [4] BioNumbers (BNID 108683), “Viscosity of the sap (typically $\sim 5\times$ water),” Harvard Medical School.
- [5] Penn State Extension, “Spotted lanternflies and beekeeping,” Oct. 5, 2025.